

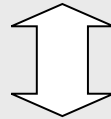
SEM-I: why and what?

Overview

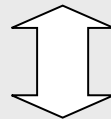
- Interfacing grammars to other systems via semantics: requirements
- What is in the SEM-I?
- SEM-I tools
- Some modest proposals ...
- SEM-I ++

Modular architecture

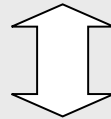
Language independent component



Meaning representation (MRS/RMRS)



Language dependent analysis/realization
(DELPH-IN grammar)



string

Semantics as interface

- Applications need to know what representations to expect / deliver:
 - transfer component for MT
 - query answering
 - information extraction, etc
- Deep/shallow integration via RMRS
 - RMRS from shallow grammars is an underspecified form of semantics from deep grammars
 - treats deep grammars as normative, so need to know their output
- Explaining what we're doing!

What must be specified

- Syntax of representation (XML)
- Formalism (MRS/RMRS)
- Naming conventions
- Attributes and values on variables
- Relations, features, constant values, variable sorts, optionality
 - `grammar' relations (e.g., **udef_q_rel**)
 - open-class relations (e.g., **_interview_v_rel**)
- Hierarchy of relations (where motivated by denotation)

Consultants were interviewed by Abrams

```
<mrs>
<var vid='h1'/>
<ep><pred>prpstn_m_rel</pred><var vid='h1'/>
<fvpair><rargname>MARG</rargname><var vid='h3'/></fvpair></ep>
<ep><pred>udef_q_rel</pred><var vid='h6'/>
<fvpair><rargname>ARG0</rargname><var vid='x4'/></fvpair>
<fvpair><rargname>RSTR</rargname><var vid='h7'/></fvpair></ep>
<ep><pred>_consultant_n_rel</pred><var vid='h9'/>
<fvpair><rargname>ARG0</rargname><var vid='x4'/></fvpair></ep>
<ep><pred>_interview_v_rel</pred><var vid='h10'/>
<fvpair><rargname>ARG0</rargname><var vid='e2'/></fvpair>
<fvpair><rargname>ARG1</rargname><var vid='x11'/></fvpair>
<fvpair><rargname>ARG2</rargname><var vid='x4'/></fvpair></ep>
<ep><pred>_by_p_cm_rel</pred><var vid='h10'/>
<fvpair><rargname>ARG0</rargname><var vid='e13'/></fvpair>
<fvpair><rargname>ARG1</rargname><var vid='u12'/></fvpair>
<fvpair><rargname>ARG2</rargname><var vid='x11'/></fvpair></ep>
<ep><pred>proper_q_rel</pred><var vid='h14'/>
<fvpair><rargname>ARG0</rargname><var vid='x11'/></fvpair>
<fvpair><rargname>RSTR</rargname><var vid='h15'/></fvpair></ep>
<ep><pred>named_rel</pred><var vid='h17'/>
<fvpair><rargname>ARG0</rargname><var vid='x11'/></fvpair>
<fvpair><rargname>CARG</rargname><constant>abrams</constant></fvpair></ep>
<hcons hreln='qeq'><hi><var vid='h3'/></hi><lo><var vid='h10'/></lo></hcons>
<hcons hreln='qeq'><hi><var vid='h7'/></hi><lo><var vid='h9'/></lo></hcons>
<hcons hreln='qeq'><hi><var vid='h15'/></hi><lo><var vid='h17'/></lo></hcons>
</mrs>
```

Some issues

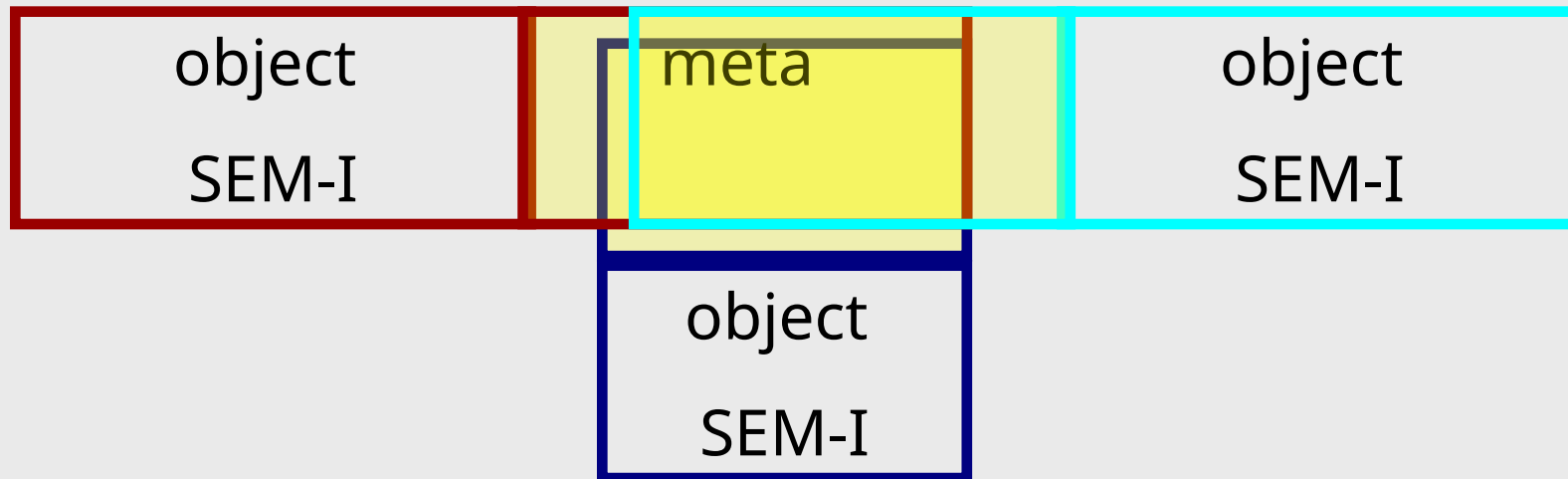
- Specification/documentation:
 - treatment of bare plural, message relations
 - defining when such relations are present
 - arity and correspondence of arguments for **_interview_v_rel** etc
- 'unwanted' predicates such as **_by_p_cm_rel** (some of these are going/gone – can all be avoided?)
- qeqs etc – can be ignored for analysis for some applications, not for realisation (currently)
- changes to grammars: e.g., message relations?

SEM-I: semantic interface

- Formal level: MRS/RMRS syntax and semantics, naming conventions (`_lemma_POS[_sense]`)
- Meta-level: variable feature values; manually specified `grammar' relations
 - **undef_q_rel** (construction)
 - **named_rel**, **proper_q_rel** (`fixed' lexical relations)
- Object-level (e.g., **_consultant_n_rel**)

SEM-I and grammars

- Object levels SEM-Is are auto-generated and distinct for each grammar
- Meta-level SEM-Is should be (partially) shared



SEM-I functionality

- Offline
 - Definition of `correct' (R)MRS for developers
 - Documentation
 - Checking of test-suites
- Online
 - SEM-I plus lexical link used in lexical lookup phase of generation (already)
 - rejection of invalid (R)MRSs (input to generator, deep/shallow integration)
 - patching up input to generation, fixing up output from parser

SEM-I: implementation (current and planned)

- Database of relations, features, value sorts, optionality:
 - Meta-level: plan to generate from grammars, with manual identification of relations (some relations are grammar-internal, see later) and manual documentation
 - Object-level: auto-generated from lexical entries in deep grammars (current version is based on generator code – optionality not there yet)
- Semantic test suite exemplifying grammar relations (partial for ERG, in progress for other grammars)

SEM-I development

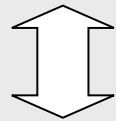
- SEM-I development must be incremental
- SEM-I eventually forms the `API': stable, changes negotiated.
 - Shared meta-level SEM-I is presumably part of Matrix, but negotiated with consumers
 - Management needs to be worked out
- Grammar writers need flexibility to hide things, make changes: SEM-I only constrains the external view
 - BUT: automate production of SEM-I from grammars as much as possible
- Documentation needs to be automated as much as possible: documentation by example

Interface

- External representation: $(R)MRS_{SEM-I}$
 - public, documented
 - reasonably stable
- Internal representation
 - mapping to feature structures (MRS_{FS})
 - MRS_{SEM-I} to MRS_{FS} mapping needed anyway, but may have to go via $MRS_{INTERNAL}$ to MRS_{FS} mapping
 - distinctions between relations which are irrelevant for denotation are hidden: only some relations are public
 - e.g., 'selected for' relations are internal only
- External/Internal inter-conversion
 - e.g., internal-only relation automatically converted to supertype in output
- BUT: want to minimize the discrepancies
 - relation hierarchies in SEM-I consistent with grammar hierarchies

Architecture with indirection

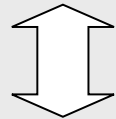
External LF (defined by SEM-I)



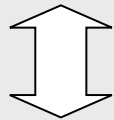
Internal LF



bidirectional
mapping

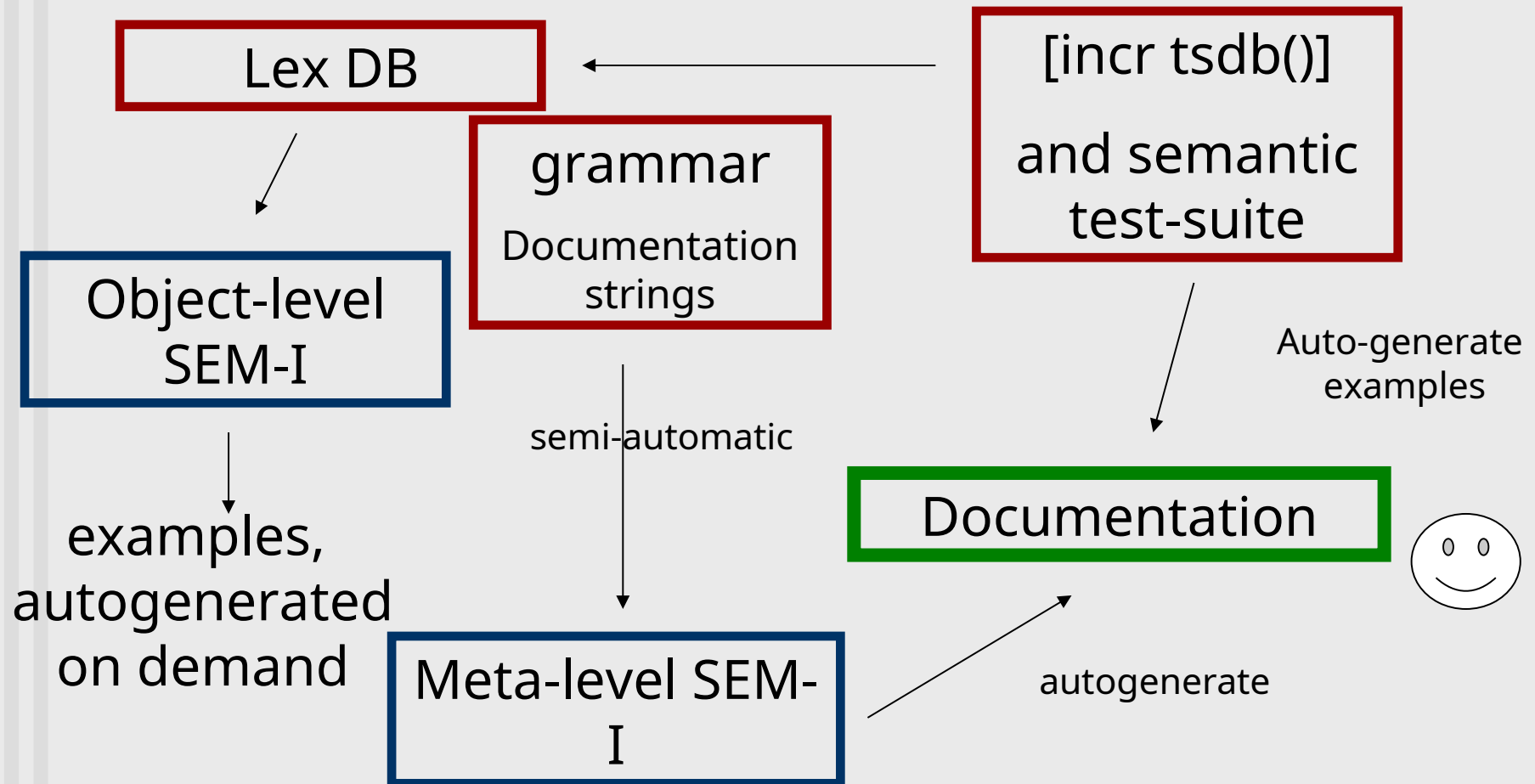


parser/generator



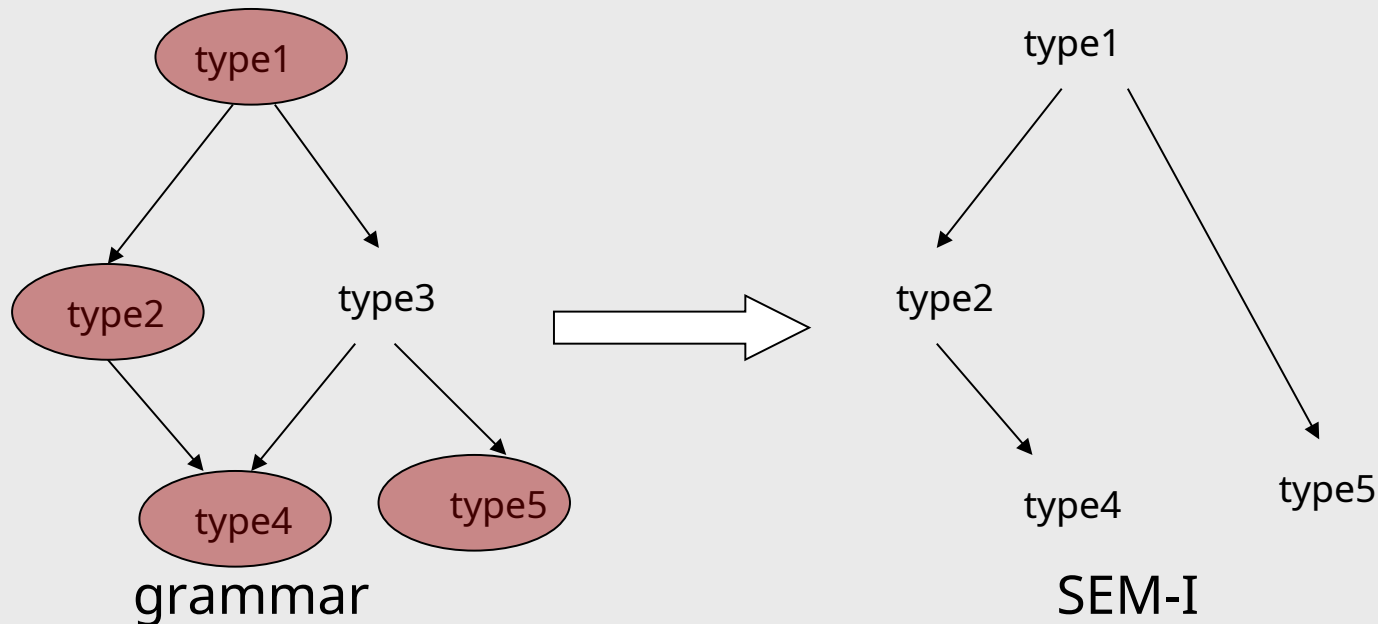
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Semi-automated documentation



Hierarchies

- Type hierarchies of relations in grammars are not there to support inference
- GLB condition not needed for SEM-I
- Proposal: basic SEM-I hierarchy of grammar relations derived automatically from grammar type hierarchy plus marking of relations as in SEM-I. (Possibly augmented in SEM-I ++, see later)



Proposals

- Documentation on wiki, mailing list for SEM-I developers and consumers
- MRS code to support particular TFS encoding of MRSs and enforce naming conventions, simplifying basic MRSFs to MRS mapping and making grammars more consistent
- Allow substantive MRS_{INTERNAL} to MRS_{SEM-I} mapping (via transfer rule mechanism), but hope to keep this minimal since it hinders deep/shallow integration.
- Agreed procedure for adding/changing variable features and values
- Inventory of grammar predicates: extensions/changes by grammar developers require notification and documentation

Change protocol (initial proposal)

A developer (grammar developer or software developer) implementing a change which will affect the SEM-I must follow the protocol:

- Consultation (meta-SEM-I only). Proposed changes to the meta-SEMI-I must be discussed on the mailing list.
- Notification. All changes to the SEM-I (meta and object) must be posted on the website.
- A script for conversion from new to old version must be posted (unless an incompatible change is agreed by the list members)
- Testing. For each grammar, there will be a semantic test suite, with agreed SEM-I output (for a specified reading). All changes to a grammar must be validated against the corresponding test-suite. All software changes must be validated against **all** test-suites. The conversion script must also be validated.
- Commit changes.

Applications and the SEM-I

- Application code will be isolated from grammar changes
 - MT: semantic transfer – mapping from one SEM-I to another
 - IE: mapping from SEM-I to template (often ignoring much of the detail in the original MRS)
 - QA: matching RMRSs: SEM-I hierarchy used for compatibility tests (also SEMI ++)

SEM-I++ (aka Floyd)

- SEM-I++ is not built by grammar developers, depends on SEM-I, not grammars
- More semantics, domain-independent, shared between applications
- Might include:
 - Definitions of grammar relations and closed-class relations to support inference
 - Mapping to external resources (e.g., WordNet and FrameNet)
 - Enriched hierarchies
 - Word classes
 - word classes could support a richer encoding of thematic role e.g., experiencer-stimulus
psych verbs map ARG1 to EXP and ARG2 to STIM
- Plan is to support specification of SEM-I++ in some version of OWL
- SEM-I++ information is additional to grammars but DELPH-IN community may agree to support it